SAMPLE

Great Observatories: The Subaru Telescope

Japan's Giant Eye

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Introduction

On the summit of Mauna Kea, beside the Hawaiian volcano's ring of giant domes, stands a cylindrical enclosure taller than a ten-story building, and inside it hangs the largest single piece of optical glass Japan has ever placed under the stars: the Subaru Telescope, the National Astronomical Observatory of Japan's 8.2-meter flagship, the instrument whose wide-field eyes have counted the universe's earliest galaxies, imaged planets around other suns, and mapped the dark energy's fingerprints across a third of the night sky. The telescope, first lit in January 1999 and named for the Pleiades, is this volume's subject: its conception and construction, its optics and instruments, its surveys and discoveries, its place on the world's most crowded summit, and its future

in the era of the thirty-meter giants.

The route is the series' standard one: the discovery and the building's history; the understanding of what the machine is and does; the characteristics, the science, and the observing craft; the famous features; the context of the sibling instruments; the modern understanding of what the telescope has settled; the research programs; the significant discoveries; the challenges and the mysteries; the future directions; a page of curiosities; and the closing verdict, the claims verified against the published record and the siblings' stories deliberately left to their own volumes.

What carries the story is a design choice: the single mirror against the segmented swarm. While its neighbors on the summit built their primaries from mosaics of hexagons, Subaru chose one huge, thin, meniscus-shaped piece of glass, held in shape by hundreds of computer-controlled fingers, and behind that mirror's wide field it mounted the survey cameras that made the telescope the world's finest galaxy-counter for a decade, the giant glass turned to the sky's biggest counting problems, the choice being the telescope's identity.

A paragraph on the instrument's standing in its nation's story belongs at the door, because Subaru is more than a telescope to the community that built it: it is Japan's first giant, the instrument that carried a patient national astronomy program into the world's first rank. Before the 1990s, Japanese observational astronomy worked with good but modest instruments at home, under sea-level skies and city light, and the decision to spend a flagship's budget on a summit four thousand meters above the Pacific, in a partner nation's high desert, was a national statement; the chapters that follow give the statement its technical content, the wide field the mirror's prime focus makes possible, the faint objects the summit's infrared darkness reveals, and the reader who finishes the volume will understand why the Pleiades name,

the flagship of the national observatory, fits the instrument's reach.

One more paragraph at the door belongs to the mountain, because no reader should meet the telescope without meeting its ground. The summit the instrument stands on is a high desert of red lava under a sky the modern world has almost run out of, high enough that the observatory crews work short shifts in thin air, remote enough that the winter nights freeze machinery that lowland instruments never test, and culturally freighted, the peak held sacred by the mountain's first people, whose tradition the observatories share as stewards as well as tenants; the volume's chapters treat the ground as seriously as the glass, because the telescope's science is a dividend of the place, and the place, not the machine, is what cannot be replaced.

Chapter 1: The Discovery of The Subaru Telescope

The discovery of the Subaru Telescope is the story of a national observatory's mountain expedition: the decision, the glass, and the first light.

The decision came in Japan's boom years, the late 1980s and early 1990s, when the National Astronomical Observatory of Japan, the research institute now under the National Institutes of Natural Sciences, resolved to build a world-class optical-infrared telescope of the eight-meter class, the country's first great optical instrument and its first permanent major observatory outside the home islands. The logic was plain: Japan's astronomy was strong in radio and in space, but the optical frontier, the wavelength where the galaxies and the stars yield their sharpest secrets, demanded a giant glass, and the giants were being built elsewhere, in Chile, in Arizona, on Mauna Kea, the moment being the last in which a national program could join the eight-meter race and matter.

The decade's competition deserves the decision's note: the eight-meter race of the 1990s, the VLT's four, Gemini's two, Keck's twin mosaics, the Japanese project entering the field as the fifth program and choosing the single-mirror road when the segment-makers had the momentum, the decision being the program's own, the wide-field survey identity being the payoff of the choice that looked conservative in 1995 and looked prescient in 2015, the decade's lesson being the design's: the instrument's best niche is chosen before the glass is cast.

The site search settled on Mauna Kea, the Hawaiian volcano's 4,139-meter summit being the northern hemisphere's finest seeing, the infrared-friendly cold air and the stable inversion layer, and the observatory's negotiators secured a site beside the planned Keck domes, the summit's science reserve accepting its first Japanese instrument. The cultural weight of the site, the

mountain's sacredness in the Native Hawaiian world, was acknowledged from the start, the telescope built on land leased through the summit's management arrangements, the context being one the volume's later chapters treat honestly, the mountain's modern astronomy and its indigenous meaning sharing the same thin air.

The glass was the boldest choice. Subaru's primary would be a single 8.2-meter meniscus, a quarter-tonne-per-square-meter pane only about twenty centimeters thick, among the largest single pieces of optical glass ever polished at its making, the meniscus design being the era's answer to the mirror's weight: too thin to hold its own shape, the glass would rest on hundreds of actuators, the computer-adjusted fingers pushing and pulling the

surface into focus all night long, the active optics' discipline imported from Europe's pioneers and scaled to the new size. The polishing, in Japan, took years against the single-digit-nanometer tolerance, the mirror arriving in Hawaii in 1998 after a Pacific crossing that treated the pane, worth some forty billion yen of project money in the decade's currency, like a relic.

The name deserves its own paragraph, the choice's care. Subaru is the Japanese for the Pleiades, the six-star cluster the car company also honors, and the observatory ran the naming through its own deliberation, the word's meaning being the point: the cluster of stars, the gathering, the telescope named for a conjunction rather than a person, the label reading as a wish, the instrument and its community joined under one sky-word, and the name's bilingual ease, instantly pronounced in every language, being the international instrument's quiet advantage, the telescope known everywhere by the same five syllables.

The commissioning year deserves the first light's note: the engineering nights of 1999, the optics' tuning and the instruments' first science, the observatory's announcements moving from the construction's bulletins to the astronomy's, the community's welcome of the newcomer's early results, the Orion images' publication being the debut's public face, and the dedication of 2000 being the formal ceremony, the telescope's youth having ended, the science's long adulthood having begun, the calendar being the machine's own rite of passage. The first light came in January 1999, the commissioning images of the Orion Nebula's glowing gas announcing the newcomer's sharpness, and the formal dedication followed in 2000, the telescope joining the summit's roster under its new name: Subaru, the Japanese word for the Pleiades, the six-star cluster that also names the car company, the observatory choosing the sky's own label, the telescope named for the heavens it would read.

The building deserves the discovery's closing note: the enclosure's cylinder,

forty-three meters of steel on the summit, the dome's ventilation lattice running down the cylinder's face, the building's size being the mirror's requirement plus the mountain's weather's margin, the structure's design having been tested in the wind tunnels of the construction era, the Japanese engineering's stamp being visible in the rigidity's numbers, the pointing's arcsecond-level steadiness in the gale, the building being the machine's unsung organ, the glass's overcoat and the crew's shelter at once. A paragraph on the project's arc from decision to first light, because a decade of construction deserves its own telling.

The design work began in the late 1980s, the site tests on the Hawaiian mountain ran through the same years, and the casting and polishing of the

great mirror, a thin meniscus of low-expansion glass more than eight meters across and only some twenty centimeters thick, took the heart of the decade; the parts were shipped across the Pacific, the mountain crews assembled the telescope inside its cylindrical enclosure at the summit, and the instrument saw first light at the end of the 1990s, the last of the great eight-meter generation to open and, by common account, among the trickiest to build.

The discovery chapter's closing note on the arc: the telescope's decade of construction is itself a discovery, proof that a national program can will a giant into existence on a mountain an ocean away.

A paragraph on the international settling-in, because the discovery story includes the politics of a summit. The mountain's observatories operate inside a host community with its own astronomy, ancient and living, and the Japanese flagship arrived in the 1990s at a sensitive moment, the mountain's future as a science reserve being negotiated between the host state, the universities, and the tradition's keepers; the telescope's builders committed to the mountain's stewardship as a condition of their tenancy, hiring locally, funding local programs, and practicing the restraint the summit's shared management asks. The founding chapter's honest note: the instrument's first discovery was diplomatic, a place at the table on a mountain nobody owns, and the lesson is one this series keeps meeting at every great site.

The glass's own journey deserves the discovery chapter's fullest paragraph, because the mirror's making was the project's odyssey. The blank was cast in America, in the same upstate New York works that has poured telescope glass for a century, a meniscus pour of a size the foundry had never attempted; the polishing fell to a Pennsylvania optics house that spent years figuring the shallow curve, the surface's error budget measured in tens of nanometers across eight meters, the pane riding its machines like a ship in a bespoke dry dock; and the finished glass crossed the Pacific by sea in a shock-mounted

cradle, the freight manifest treating the nation's flagship as the most delicate cargo the decade's oceans carried, the pane arriving in Hawaii to be lifted, once, up the volcano's access road to the summit, the whole journey being the observatory's favorite construction story, the glass having survived the world before it ever saw the sky.

The funding story deserves its own note, because the project's price was a national argument. The telescope's budget, in the decade's currency one of the largest civilian science expenditures Japan had made, moved through the national accounts beside the era's other great instruments, the radio arrays and the space programs, and the parliament's questions were the ones big science always faces: why a summit in a foreign ocean, why glass when the

field's future was thought to be in orbit. The observatory's answer, made in budgets and defended in hearings, was the one the later chapters vindicate: a giant glass on the world's best site is a national asset no spacecraft can replace, the ground being where the light is gathered and the decades are spent. The card for the reader: the telescope was bought twice, once with yen and once with the years of advocacy that moved the yen, the second purchase being the harder one.

The summit's construction deserves its own paragraph, because the dome was built in the mountain's own weather. The enclosure rose through the volcano's winters, the crews working the high-altitude discipline before the instrument existed, the concrete poured and cured in freezing nights, the steel lifted in wind windows, the building's own ventilation lattice tested against the same gales it would later tame; the summit's road, the access ribbon that carries everything the observatory owns, was the construction's lifeline, the pane's cradle having crawled it once. The card is the craft's context: the dome's cylinders and the telescope's mount were assembled on a desert island of rock above the clouds, the engineers living at the volcano's waist and commuting to the sky, the machine's first environment being the mountain's worst season, the building's every bolt an altitude story, the construction's decade being as much mountaineering as engineering.

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